Harm-Related Injuries Among Young People Visiting Emergency Department in Edmonton Police District

Background

Data included in this document were compiled by request for Edmonton Police Service.

If there are any questions, or if additional data would be helpful, please contact Janis Geary (janis.geary@gov.ab.ca).

Data Notes

- Population: Residents of Alberta visiting any Emergency Department in the Edmonton Police District.
- Demographic data was captured from population registries at the Ministry of Primary and Preventative Health Services.
- Emergency Department visit data came from the Ambulatory Care Classification System.
- Suicide, self-inflicted injury and accidental poisoning cases were excluded from this report.
- Please also note that results may differ depending on the inclusion and exclusion criteria used to define various conditions and by methods used for estimation. As such, these results may be different from those obtained through other sources or if analyses are repeated using different criteria or at different times.
- A single patient may have been seen by a doctor for more than one type of harm and/or violence in each visit however, only the main type reported was selected for reporting.
- If an outcome had fewer than 10 events, the outcome was not included in the data due to potential identifiability.

Definitions

- Emergency department (ED) visit data comes from the ambulatory care datasets. Violence and injury purposely inflicted were selected using the international classification of diseases version 10 (ICD-10) (see Appendix A).
- The Physician claims data (SESE or Supplemental Enhanced Service Event data) contains all of the claims assessed by Alberta Health for payment by physicians and other health care providers. Diagnosis categories were selected using the international classification of diseases version 9 (ICD-9) (see Appendix A).

Key Points

- Across all age categories, the most ED visits in Edmonton Police District due to harm and violence among young people were attributed to physical altercations and brawls (Table 1).
- Edmonton – Eastwood consistently had the highest volume (Table 4) and rate of ED visits/100,000 in Edmonton (Table 5, Figure 1).
- Few ED visits were individuals under 18 from outside of the Edmonton Corridor or who were flagged as experiencing homelessness (Tables 7 and 8).
- Most patients are discharged home from the ED, although this proportion has been decreasing (73.6% to 66.2%), and there are increases in the per cent leaving without being seen (13.3% to 20.4%) and the per cent discharged to continuing or resident care (0.9% to 3.0%) (Table 9).
- The most ED visits in people with more than three visits per were attributed to physical altercations and sexual harms (Table 13).
- While a small proportion of individuals are visiting EDs more than 3 times due to violence/harms, these individuals have a lower per cent that are discharged home (58.2% vs 70.6%), and a higher per cent that leave the ED without being seen (26.8% vs 16.5%). They also have a higher per cent of visits due to sexual harms (29.0% vs 18.6%)
- Across all age categories, people who visited the ED due to violence or harm had a higher percentage of doctor visits for mental disorders in the two years prior to their first ED visit compared to those who did not experience violence (Table 15). When looking at specific subcategories of reasons for visits, depressive disorders and psychotic conditions were more common reasons for seeing a physician, and fractures/sprains were less common (Table 16).

Data

Table 1: Per cent of types of violence for unique individuals visiting EDs in Edmonton Police District related to harm and violence, by type of violence/harm and age group, 2018-2024.

Harm/Violence Category	10-11	12-14	15-17	18-24
	N=160	N=796	N=1875	N=6191
Assault	10.6%	11.4%	21.9%	29.2%
Fight/Brawl	56.3%	39.1%	36.3%	40.7%
Firearms	0.0%	-	0.9%	1.6%
Legal intervention	0.0%	-	1.1%	2.6%
Physical Abuse	-	2.6%	1.3%	1.4%
Sexual harms	20.6%	28.1%	25.2%	15.7%
Undetermined whether unintentionally or purposely inflicted	-	14.1%	10.8%	7.2%
Maltreatment	-	-	0.8%	0.6%
Other	5.0%	2.6%	1.8%	1.1%

Table 2: Most Common Harms Leading to ED Visits Among the Two Most Frequent Age Groups in Edmonton Police District, 2018-2024

Age Group (years)	Harm/Violence Category	2018	2019	2020	2021	2022	2023	2024
15-17	Assault	57	58	56	53	50	68	69
	Fight/Brawl	120	105	68	75	74	106	132
	Legal intervention	<10	<10	<10	<10	<10	<10	<10
	Sexual harms	70	66	63	69	56	73	75
18-24	Assault	305	315	267	217	217	232	252
	Fight/Brawl	464	447	327	288	349	311	334
	Legal intervention	38	32	20	11	23	20	18
	Sexual harms	167	145	126	150	127	136	124

Table 3: The number individuals visiting different Eds in Edmonton Police District due to harm and violence, 2018-2024

Facility Name	2018	2019	2020	2021	2022	2023	2024
Royal Alexandra Hospital	431	400	333	308	312	310	313
University of Alberta total	510	452	431	494	474	513	614
University of Alberta Hospital	510	437	224	234	217	215	239
Stollery Children's Hospital*	0	15	207	260	257	298	375
Grey Nuns Community Hospital	182	210	139	128	118	115	146
Northeast Community Health Center	193	186	127	103	147	137	115
Misericordia Community Hospital	188	164	109	98	97	108	141
East Edmonton Family Care Clinic	23	32	18	16	21	25	41

* Stollery Children's Hospital was reported within University of Alberta Hospital Data prior to 2020

Table 4: Number of ED visits by LGA for Edmonton Corridor residents visiting EDs in Edmonton Police District due to harm and violence (per 100,000 person years)

Edmonton Corridor- LGA Name	2018	2019	2020	2021	2022	2023	2024
Edmonton - Eastwood	170	148	109	114	135	137	142
Edmonton - Northeast	142	117	105	80	105	94	114
Edmonton - Northgate	116	127	73	62	82	86	91
Edmonton - Bonnie Doon	100	94	90	77	80	87	104
Edmonton - Woodcroft East	93	75	75	62	65	70	79
Edmonton - West Jasper Place	86	76	59	65	54	69	72
Edmonton - Mill Woods South | Mill Woods East	68	68	62	57	50	48	78
Edmonton - Jasper Place	69	68	52	53	52	58	70
Edmonton - Mill Woods West	62	75	47	48	41	65	51
Edmonton - Castle Downs	66	61	37	53	59	47	58
Edmonton - Rutherford	46	61	34	51	47	63	63
Edmonton - Duggan	30	31	26	38	35	32	37

Edmonton - Woodcroft West	32	39	39	25	33	20	38
Edmonton - Twin Brooks	29	42	21	28	33	32	28
County Of Wetaskiwin	28	23	28	28	35	28	27
Edmonton - Abbottsfield	45	26	22	20	20	25	28
Other LGA	213	190	168	186	164	162	200

Note: 282 LGA name not reported due to missing data.

Table 5: Rate of ED visits by LGA for Edmonton Corridor residents visiting EDs in Edmonton Police District due to harm and violence (per 100,000 person years)

Edmonton Corridor- LGA Name	Rate of ED Visits for Violence or Harms/100,000 person years						
	2018	2019	2020	2021	2022	2023	2024
Edmonton - Eastwood	1550	1342	1018	1093	1214	1175	1127
Edmonton - Woodcroft East	1166	929	925	773	773	772	814
Edmonton - Jasper Place	844	834	645	674	654	702	798
Edmonton - Northeast	846	678	591	442	559	473	531
Edmonton - Woodcroft West	612	732	718	460	590	332	596
Edmonton - Bonnie Doon	614	580	559	512	505	523	595
Edmonton - Northgate	726	788	450	384	498	498	501
Edmonton - Mill Woods West	587	709	444	458	393	598	431
Edmonton - Duggan	382	392	326	495	443	396	431
County Of Wetaskiwin	383	314	380	377	472	371	351
Edmonton - Castle Downs	485	434	258	362	391	296	349
Edmonton - West Jasper Place	470	400	299	316	253	304	298
Edmonton - Mill Woods South | Mill Woods East	405	394	346	312	261	236	335
Edmonton - Rutherford	294	339	173	235	196	231	204
Edmonton - Twin Brooks	198	283	141	187	217	207	174

Note: 282 LGA name not reported due to missing data.

Figure 1: Rate of ED visits by LGA for Edmonton Corridor residents visiting EDs in Edmonton Police District due to harm and violence (per 100,000 person years)

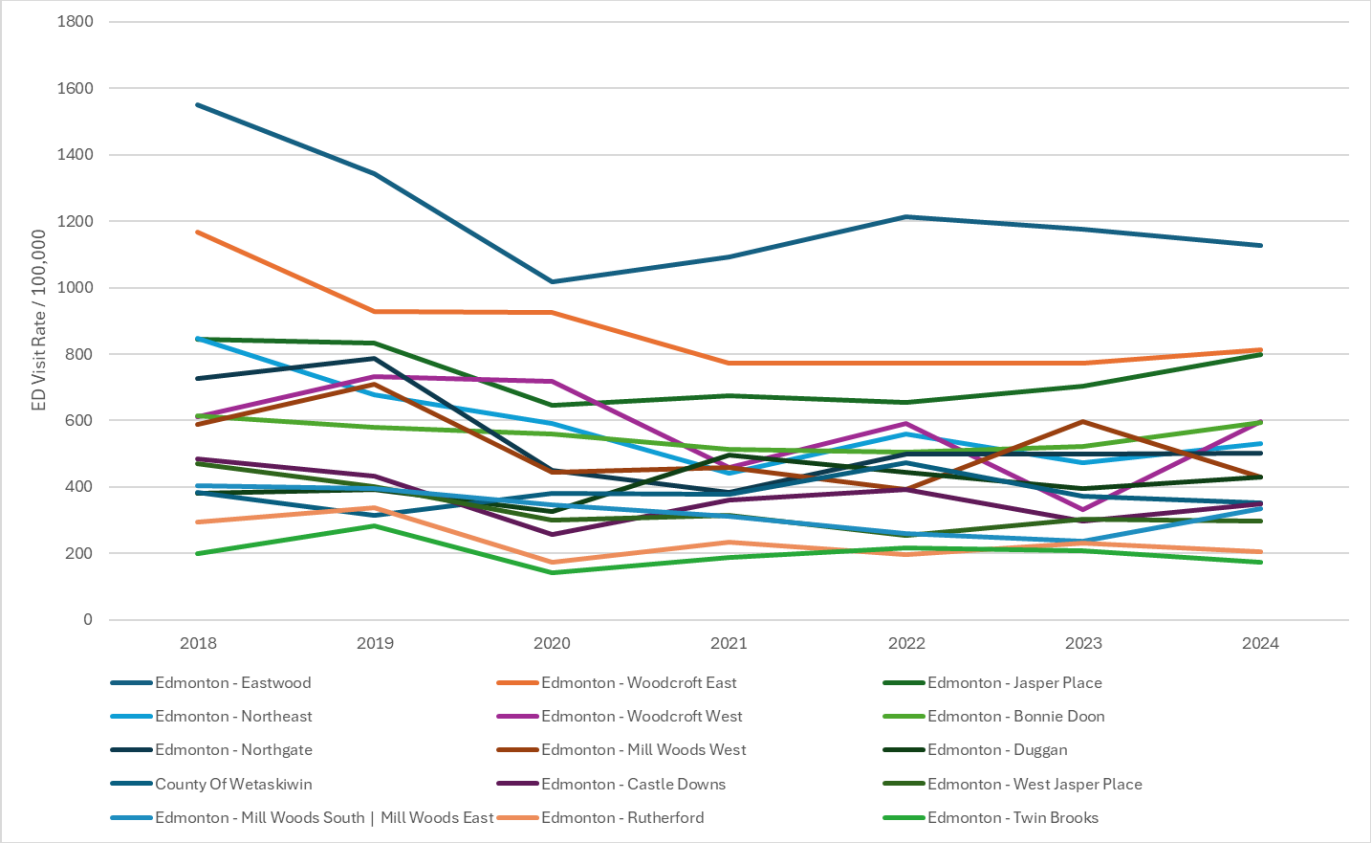

Table 6: Number of ED visits by LGA for Edmonton Corridor residents visiting EDs in Edmonton Police District due to harm and violence (per 100,000 person years), by age group, 2018-2024

Edmonton Corridor- LGA Name	10-11	12-14	15-17	18-24
Edmonton - Eastwood	15	42	146	752
Edmonton - Northeast	11	93	163	490
Edmonton - Northgate	13	57	152	415
Edmonton - Bonnie Doon	<10	35	129	463
Edmonton - Woodcroft East	10	44	83	382
Edmonton - West Jasper Place	10	41	124	306
Edmonton - Mill Woods South | Mill Woods East	<10	48	103	273
Edmonton - Jasper Place	<10	30	78	309
Edmonton - Mill Woods West	10	50	85	244
Edmonton - Castle Downs	<10	54	96	226
Edmonton - Rutherford	15	39	96	215
Edmonton - Duggan	<10	15	55	153
Edmonton - Woodcroft West	<10	21	34	164
Edmonton - Twin Brooks	<10	26	48	138
County Of Wetaskiwin	<10	10	29	157
Edmonton - Abbottsfield	<10	15	46	120
Other LGA within the Edmonton Corridor	27	140	314	802

Note: 282 LGA name not reported due to missing data.

Table 7: Number of ED visits by LGA for non-Edmonton Corridor residents visiting EDs in Edmonton Police District due to harm and violence (per 100,000 person years), by age group, 2018-2024

LGA of residence	10-11	12-14	15-17	18-24
High Level	<10	<10	<10	35
High Prairie		<10	10	29
Wabasca	<10	<10	<10	25
Rocky Mountain House			<10	21
Fort McMurray		<10	<10	17
Slave Lake		<10	<10	19
Peace River		<10	<10	13
Grande Prairie			<10	14
Red Deer - East		<10		<10
Wood Buffalo Excluding Fort McMurray		<10		10
Other	<10	14	22	147

Table 8: Number of Homeless Individuals Visiting EDs in Edmonton Police District due to harm and violence, by year and age group, 2018-2024

Year	Age group (year)			
	10-11	12-14	15-17	18-24
2018	0	<10	18	80
2019	0	<10	<10	99
2020	0	0	20	71
2021	0	0	13	79
2022	0	0	<10	64
2023	0	<10	19	84
2024	<10	0	22	74

Table 9: Per cent of ED visits due to harm and violence with different dispositions in Edmonton Police District, by year, 2018-2024 (per cents for counts lower than 10 are suppressed)

Disposition	2018	2019	2020	2021	2022	2023	2024
	N=1785	N=1690	N=1348	N=1309	N=1356	N=1442	N=1654
Discharged Home	73.6%	73.4%	73.8%	72.8%	67.4%	66.5%	66.2%
Left ED Without Being Seen / AMA	13.3%	14.6%	12.3%	13.3%	19.8%	21.7%	20.4%
Admitted (Inpatient care)	10.1%	9.5%	10.7%	10.2%	8.4%	8.3%	8.9%
Discharged to Continuing/Residential Care	0.9%	0.9%	1.4%	1.7%	2.3%	2.1%	3.0%
Correctional Facility: Transfer to jail or halfway house	0.8%	1.3%	1.0%	1.0%	1.3%	0.7%	1.3%
Intra-Facility Transfer	-	-	-	-	0.7%	-	-
Other / Unknown	1.0%	0%	0%	0%	0%	0%	0%
Died	-	-	-	-	-	0%	-

Table 10: Per cent of ED visits due to harm and violence with different dispositions in Edmonton Police District, by Age Group, 2018-2024 (per cents for counts lower than 10 are suppressed)

Disposition	Age group			
	10-11	12-14	15-17	18-24
	N=169	N=911	N=2254	N=7250
Discharged Home	85.8%	79.3%	72.8%	68.5%
Left ED Without Being Seen / AMA	7.1%	9.5%	13.7%	18.4%
Admitted (Inpatient care)	-	5.8%	8.5%	10.4%
Discharged to Continuing/Residential Care	-	4.5%	3.7%	0.7%
Correctional Facility: Transfer to jail or halfway house	0%	-	0.6%	1.3%
Intra-Facility Transfer	-	-	0.4%	0.4%
Other / Unknown	0%	0%	-	0.2%
Died	0%	0%	-	-

Table 11: Per cent of individuals with different frequencies of ED use due to harm and violence in Edmonton Police District, 2018-2024

Year	N	Repeat Visit			
		Unique Individuals With 1 Visit	Unique Individuals With 2 Visits	Unique Individuals With 3 Visits	Unique Individuals With ≥ 4 Visits
2018	1527	87.8%	9.3%	2.0%	0.9%
2019	1444	87.1%	10.5%	1.7%	0.8%
2020	1157	87.7%	9.6%	1.7%	1.0%
2021	1147	88.8%	9.4%	1.2%	0.6%
2022	1169	87.6%	10.2%	1.5%	0.7%
2023	1208	85.5%	11.8%	1.5%	1.2%
2024	1370	85.8%	11.4%	1.6%	1.2%

Table 12: ED Dispositions due to harm and violence among unique individuals with more than three visits per year, 2018-2024

Disposition	Count	Per cent
Discharged Home	502	58.2%
Left ED Without Being Seen / AMA	231	26.8%
Admitted (Inpatient care)	82	9.5%
Discharged to Continuing/Residential Care	33	3.8%
Correctional Facility: Transfer to jail or halfway house	<10	-
Other / Unknown	<10	-
Intra-Facility Transfer	<10	-

Classification: Protected A

Table 13: Type of violence/harm experienced by individuals with more than three ED visits per year in Edmonton Police District due to harm and violence, 2018-2024

Type of Violence/Harm	Count	Per Cent of ED visits
Fight/Brawl	294	34.1%
Sexual harms	250	29.0%
Assault	220	25.5%
Undetermined whether unintentionally or purposely inflicted	55	6.4%
Legal intervention	18	2.1%
other	10	1.2%
Physical Abuse	10	1.2%
Firearms	<10	-
maltreatment	<10	-

Table 14: Number of patients with no physician claims* 2 years before first ED visit in Edmonton Police District, by age group, 2018-2024

Age Group (years)	Unique patients (2018-2024)	No physician claims 2 years before first ED Visit (n)	No physician claims 2 years before first ED Visit (%)
10-14	920	10	1.09%
15-17	1655	29	1.75%
18-24	5236	207	3.95%

*Note that this includes all physician claims, and not just primary care.

Table 15: Per cent of visits for top reasons (large illness categories) for visiting a doctor 2 years before the first ED visit among young people in Edmonton Police District, 2018-2024

Broad Diagnosis Category	% of billing claim codes for different reasons among those who visited an ED for violence/harm				% of billing claim codes for different reasons among those who *did not* visit an ED for violence/harm			
	10-11	12-14	15-17	18-24	10-11	12-14	15-17	18-24
Mental Disorders	41%	55%	49%	46%	15%	19%	25%	24%
Symptoms not linked to a specific diagnosis	14%	9%	8%	9%	18%	16%	14%	14%
Injury and Poisoning	11%	11%	13%	12%	11%	13%	11%	8%
Other Disease	8%	6%	7%	6%	12%	13%	12%	10%
Supplementary Classification	8%	7%	7%	7%	9%	8%	8%	11%
Diseases of the Respiratory System	6%	3%	3%	3%	9%	7%	6%	5%
Diseases of the Nervous System	5%	2%	2%	2%	6%	6%	5%	4%
Infectious and Parasitic Diseases	2%	2%	2%	2%	6%	4%	3%	3%
Diseases of the Skin	2%	2%	2%	2%	3%	3%	4%	4%
Diseases of the Genitourinary System	1%	1%	2%	2%	2%	2%	2%	4%

Table 16: Per cent of visits for top reasons (illness subcategories) for visiting a doctor 2 years before the first ED visit among young people, 2018-2024

Diagnosis Sub-Category	% of billing claim codes for different reasons among those who visited an ED for violence/harm				% of billing claim codes for different reasons among those who *did not* visit an ED for violence/harm			
	10-11	12-14	15-17	18-24	10-11	12-14	15-17	18-24
Depressive disorder, not elsewhere classified	11%	27%	21%	16%	3%	11%	19%	16%
Psychotic conditions (e.g., Schizophrenic, Affective psychoses)	4%	12%	21%	34%	4%	5%	10%	28%
Hyperkinetic syndrome of childhood	36%	9%	5%	1%	23%	12%	7%	2%
Conduct disorder (CD)	10%	16%	12%	3%	8%	10%	10%	3%
Neurotic disorders (e.g. Anxiety)	7%	10%	9%	6%	9%	9%	11%	10%
Fracture of skull, spine and trunk, upper limb, lower limb, dislocation	5%	3%	3%	3%	18%	17%	9%	6%
Sprains and strains of joints and adjacent muscles	4%	2%	2%	2%	10%	12%	10%	7%
Intracranial injury, Internal injury, Injury to blood vessels	5%	2%	3%	2%	5%	4%	4%	3%
Open wounds	3%	2%	3%	3%	5%	4%	3%	3%
Poisoning by drugs, medicinal and biological substances	1%	5%	5%	5%	<1%	2%	3%	3%
Contusions	3%	1%	1%	1%	4%	3%	2%	1%
Adjustment reaction	1%	3%	3%	1%	1%	1%	2%	2%

Appendix A

Emergency Department visits

Harm Indicators

- At least one of the following ICD 10 CA codes were present in any diagnostic field for the visit to be included for violence injuries:
    - T74.1: Physical Abuse– Physical abuse
    - Z61.6: Physical Abuse– Problems related to alleged physical abuse of child
    - Y07.0: Maltreatment – By spouse or partner
    - Y07.1: Maltreatment – By Parent
    - Y07.2: Maltreatment – By acquaintance or friend
    - Y07.3: Maltreatment – By official authorities
    - Y07.8: Maltreatment – By other specified persons
    - Y07.9: Maltreatment – By other unspecified persons
    - T748: Maltreatment – Other maltreatment syndromes
    - T749: Maltreatment – Maltreatment syndrome, unspecified
    - X85: Assault – Assault by drugs, medicaments and biological substances
    - X86: Assault – Assault by corrosive substance
    - X87: Assault – Assault by pesticides
    - X88: Assault – Assault by gases and vapors
    - X89: Assault – Assault by other specified chemicals and noxious substances
    - X90: Assault – Assault by unspecified chemical or noxious substances
    - X91: Assault – Assault by hanging strangulation and suffocation
    - X92: Assault – Assault by drowning and submersion
    - X96: Assault – Assault by explosive material
    - X97: Assault – Assault by smoke, fire and flames
    - X98: Assault – Assault by steam, hot vapors and hot objects
    - X99: Assault – Assault by sharp object
    - Y00: Assault – Assault by blunt object
    - Y01: Assault – Assault by pushing from high place
    - Y02: Assault – Assault by pushing or placing victim before moving object
    - Y03: Assault – Assault by crashing of motor vehicle
    - Y04: Fight/Brawl – Assault by bodily force
    - Y08: Assault – Assault by other specified means
    - Y09: Assault – Assault by other unspecified means
    - Y87.1: Assault – Sequalae of assault
    - X93: Firearms – Assault by handgun discharge
    - X94: Firearms – Assault by rifle, shotgun and larger firearm discharge
    - X95.00: Firearms – Assault by other and unspecified firearm discharge
    - X95.01: Firearms
    - X95.08: Firearms
    - X95.09: Firearms
    - Y10-Y34: Undetermined whether unintentionally or purposely inflicted
    - Y35-Y36: legal intervention
    - Y89.0: legal intervention
    - Y06.0: Neglect/Abandonment – By spouse or partner
    - Y06.1: Neglect/Abandonment – By parent
    - Y06.2: Neglect/Abandonment – By acquaintance or friend
    - Y06.8: Neglect/Abandonment – By other specified persons
    - Y06.9: Neglect/Abandonment – By unspecified person
    - T74.0: Neglect/Abandonment – Neglect or abandonment
    - Z624: Neglect/Abandonment – Emotional neglect of child
    - Z625: Neglect/Abandonment – Other problems related to neglect in upbringing
    - T743: Psychological abuse
    - Y05: Sexual Harm– Sexual assault by bodily force
    - T74.2: Sexual Harm– Sexual Abuse
    - Z04.4: Sexual Harm– Examination and observation following alleged rape and seduction
    - Z04.50: Sexual Harm– Examination and observation following alleged adult sexual and physical abuse
    - Z04.51: Sexual Harm– Examination and observation following alleged child sexual and physical abuse

- Z61.4: Sexual Harm– Problems related to alleged sexual abuse of child by person within primary support group
- Z61.5: Sexual Harm– Problems related to alleged sexual abuse of child by person outside primary support group
- Z914: Other– Personal history of psychological trauma, not elsewhere classified
- T730: Other– Effects of hunger
- Z620: Other– Inadequate parental supervision and control
- Z621: Other– Parental overprotection
- Z626: Other–Inappropriate parental pressure and other abnormal qualities of upbringing
- Z628: Other– Other specified problems related to upbringing
- Z0458: Other– Examination and observation following other inflicted injury

Physician Claim Indicators

- At least one of the following ICD 9 codes were present in any diagnostic field for the claims to be included:
  - Infectious And Parasitic Diseases: 001-139
  - Neoplasms: 140-239
  - Endocrine, Nutritional and Metabolic Diseases, And Immunity Disorders: 240-279
  - Diseases Of the Blood and Blood-Forming Organs: 280-289
  - Mental Disorders: 290-319
  - Diseases Of the Nervous System and Sense Organs: 320-389
  - Diseases Of the Circulatory System: 390-459
  - Diseases Of the Respiratory System: 460-519
  - Diseases Of the Digestive System: 520-579
  - Diseases Of the Genitourinary System: 580-629
  - Complications Of Pregnancy, Childbirth, And the Puerperium: 630-679
  - Diseases Of the Skin and Subcutaneous Tissue: 680-709
  - Diseases Of the Musculoskeletal System and Connective Tissue: 710-739
  - Congenital Anomalies: 740-759
  - Certain Conditions Originating in The Perinatal Period: 760-779
  - Symptoms, Signs, And Ill-Defined Conditions: 780-799
  - Injury And Poisoning: 800-999
  - Supplementary Classification of Factors Influencing Health Status and Contact with Health Services: V01-V99
    - Persons With Potential Health Hazards Related to Communicable Diseases (V01-V07)
    - Persons With Potential Health Hazards Related to Personal and Family History (V10-V19)
    - Persons Encountering Health Services in Circumstances Related to Reproduction and Development (V20-V28)
    - Healthy Liveborn Infants According to Type of Birth (V30-V39)
    - Persons With a Condition Influencing Their Health Status (V40-V49)
    - Persons Encountering Health Services for Specific Procedures and Aftercare (V50-V59)
    - Persons Encountering Health Services in Other Circumstances (V60-V68)
    - Persons Without Reported Diagnosis Encountered During Examination and Investigation Of Individuals and Populations (V70-82)
    - Genetic And Other Risk Factors (V83–V89)
    - Miscellaneous Categories (V90-V99)